Jordan Tran

jordantran092@gmail.com | Website Portfolio | GitHub | 647-323-2782

Summary

Computer Science student with hands-on experience building **full-stack applications** across **TypeScript/Next.js** and **Java/Spring Boot**, including **RESTful APIs, relational databases, authentication,** and **cloud deployment**. Experienced in designing and implementing **layered architectures** and developing reliable software through **TDD** and **automated testing**. Built production-deployed applications using **PostgreSQL, Docker, AWS,** and **Vercel**.

Education

Honours Bachelor of Science in Computer Science, York University, Toronto — Expected Oct 2027
Relevant Courses: Data Structures, Databases, Algorithms, Software Design

Relevant Projects

Hierarchical Document Workspace App | **Demo** | **Code** | **Live Site**

- **Built and deployed** a **full-stack hierarchical document workspace app** via **Next.js, React, TypeScript** & **Tailwind** with **authentication, full-text search,** block-based rich-text editing & nested page structure
- Designed a **layered architecture** separating **Server Actions, service-layer business logic, Prisma ORM,** and React server/client components for **maintainability** via **separation of concerns**
- Implemented **session-based authentication** (BetterAuth) and **ownership-based authorization** across page operations to enforce **secure, private data access**
- Developed **full-text search** via **GIN indexing, pagination**, debounced search, highlighted matches, relevance ranking & **stored vectors** to **avoid recomputation** for **SQL query performance**
- Integrated **BlockNote** with **custom blocks and menus,** JSON document persistence, breadcrumb page navigation**, debounced autosave** to **reduce unnecessary database writes**
- Designed **DB schema** & integrated **PostgreSQL** via **Prisma** for **streamlined data access**, **deploying** the production application on **Vercel**

Bank Mobile App | **Demo** | **Code**

- **Built and deployed** a **RESTful Java full-stack banking system** via **Spring Boot** & **PostgreSQL**, using a **layered architecture** (controller, service, repository) for **maintainability**
- Designed **DB schema** & integrated **PostgreSQL** via **Hibernate ORM** for **streamlined data access**
- Built **client UI** in **Android Studio** & integrated **5 REST endpoints** by **analyzing user interface flow**, enabling core banking operations (account creation, deposit, withdraw, transfer, statement generation)
- Applied **TDD, JUnit, Mockito,** and **MockMvc**, covering **15 automated unit, controller** and **service integration-level tests** for **code reliability** & **efficient debugging** through **early defect detection**
- Integrated **business workflow validation** in a **core service layer** before delegating operations, **preventing invalid operations** from propagating deeper into the system
- **Dockerized** Spring Boot API & **deployed** to **AWS Lightsail,** for **mobile app access** to **REST API**

Driving Simulation | **Demo** | **Code**

- Built a **Python-based** driving simulation to model user progression on a virtual road
- Utilized **PyGame API** to develop **application logic & UI elements** (e.g. vehicles, live progress bar)
- Designed **dynamic generation** of **AI traffic** & **rule-based movement logic** with **attention to detail,** producing unpredictable collision scenarios that raise the game's challenge
- Developed **3 termination conditions to handle state transitions** involving bounding box collision detection of user & AI, screen boundary violations, and finish-line completion logic

Technical Skills

Languages: Java, Python, TypeScript, SQL, HTML, CSS, Bash, C

Backend: Spring Boot, Next.js, Node.js, Hibernate ORM, Prisma, BetterAuth, JUnit, MockMvc, Mockito

Frontend: React, Tailwind, Bootstrap, Figma

DevOps/Cloud: Docker, AWS Lightsail, Vercel, Azure

Databases & Other: PostgreSQL, SQL Server, Git, GitHub, Android Studio

Volunteering

Orientation Logistics Volunteer — **August 2019**
University of Toronto

- Assembled **8000+ student orientation kits** with student supplies in a **fast-paced, deadline-driven environment** through **constant communication** and **teamwork** with **100+ volunteers**
- Collaborated with **volunteers** to manage **5000+ by-products** from packing to maintain an **organized workspace**